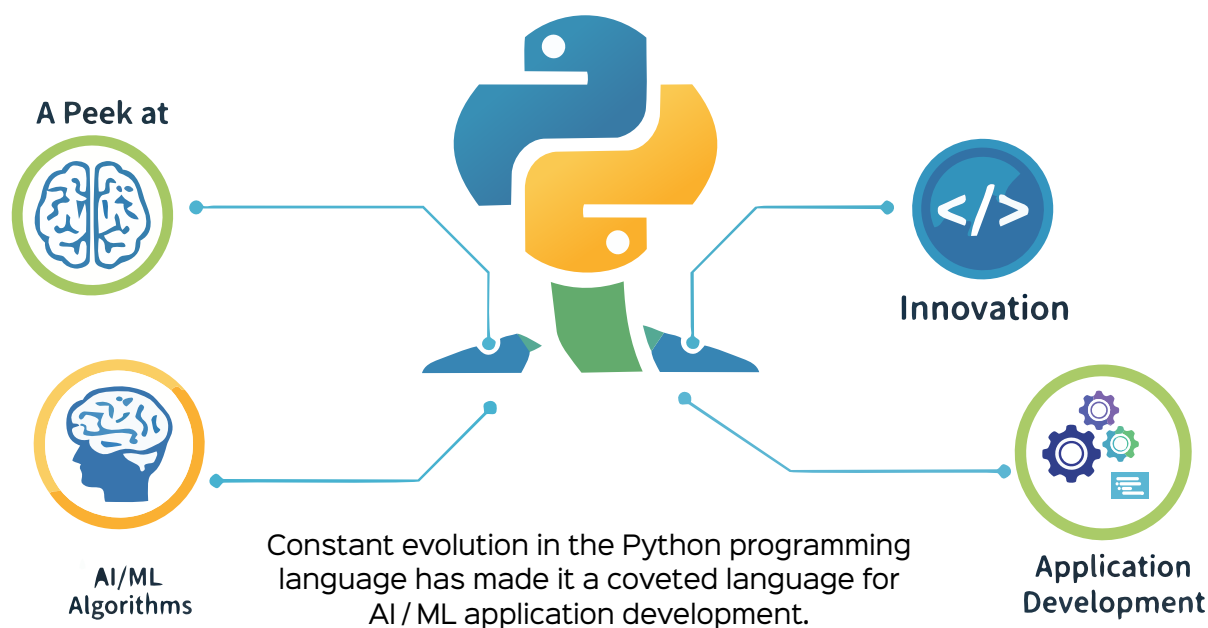


A Peek at Python's Prowess in AI/ML Application Development



Python is perhaps the most popular language for AI and ML application development. Its rich libraries and tooling streamline everything from rapid prototyping to production deployment, while built-in data structures and libraries make preprocessing, exploration, and manipulation of datasets straightforward and efficient.

Python streamlines rapid AI/ML application development, automates machine learning workflows, and supports innovation and integration with cloud platforms and IoT for AI/ML applications. For developers, Python combines productivity, community support, and production-grade capabilities, making it the natural choice for building, training, and deploying intelligent systems.

Python continues to evolve, yet many developers still rely on the techniques they first learned. Those legacy approaches are not inherently wrong, but can hide newer, more efficient patterns, libraries, and idioms that improve performance, readability, and maintainability. Regularly updating skills, adopting modern libraries, idiomatic constructs, and contemporary tooling reduces bugs, speeds development, and smooths the path from prototype to production.

Python adoption in the industry

Python powers over half of global data science projects and holds a commanding lead in industry adoption. The TIOBE

Programming Community Index (December 2025) shows Python ranked #1 with a 23.64% share, ahead of C (10.11%) and C++ (8.95%), making it the most popular programming language for AI and ML. Over 51% of all data science projects worldwide are built with Python. More than 1.25 million open positions globally require Python skills, with senior ML developers earning up to \$200,000 in the US.

Python powered 41% of all AI-generated code in 2025, with 256 billion lines written by AI assistants like GitHub Copilot. The Python IDE market was valued at US\$ 559 million in 2025 and is expected to grow at 8.1% CAGR through 2033.

Netflix's recommendation algorithm processes over 1 billion hours of content using Python-based machine learning models.

The characteristics of Python are:

- Processed at runtime by the interpreter. No need to compile the program before executing it.
- Can run in an interactive mode, i.e., program can be written in a command line shell, which gives immediate output for each statement.
- Cross-platform -- works on different platforms like Windows, Linux, Mac, etc.
- Easy to learn and use; developer-friendly.
- Easy to read and understand; the syntax of Python is like English statements.